

Supervision Intensity-Guided Adaptive One-to-Many Matching for Long-Tailed Object Detection

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ACM Reference Format:

Yandan Wang, Jiangyan Chen, Yinglong Ma, Chenqi Guo, Jiaqi Ma, and Weiming Dong. 2026. Supervision Intensity-Guided Adaptive One-to-Many Matching for Long-Tailed Object Detection. In *Proceedings of the 35th ACM International Conference on Multimedia (MM '26)*, November 10–14, 2026, Rio de Janeiro, Brazil. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3767308.3834955>

A Appendix

A.1 Sensitivity to the Trade-off Coefficient β

We study the sensitivity of the trade-off coefficient β in SI. In Eq. (4) in main paper, β controls the balance between the boundary-aware reliability term $U_m(r)$ and the retained positive quantity term $N_m(r)$.

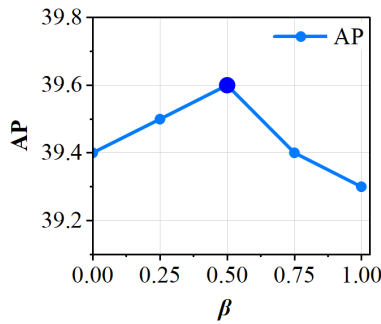


Figure 1: Sensitivity analysis of the trade-off coefficient β in the supervision intensity on LVIS.



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MM '26, Rio de Janeiro, Brazil.

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ACM ISBN 979-8-4007-2213-4/2026/11
<https://doi.org/10.1145/3767308.3834955>

As shown in Fig. 1, AP first increases and then decreases as β grows, reaching its optimum at 0.5. When the value is too small, positive-set construction becomes overly conservative and may discard useful supervision. The best result at 0.5 suggests that long-tailed detection requires a proper balance between supervision coverage and positive-set reliability. Therefore, 0.5 is used in all experiments. A pronounced performance drop appears mainly at the more extreme settings, namely 0.75 and 1.0, whereas the gap between 0.25 and 0.5 remains small. This indicates that noticeable degradation occurs only when positive-sample selection shifts too far from reliability-oriented filtering toward quantity-oriented retention, which substantially weakens supervision quality.

A.2 Model Complexity Analysis

As shown in Table 1, we report the comparison of AP, FLOPs, and parameter count on the LVIS v1.0 validation set.

Table 1: Comparison of model efficiency on LVIS v1.0 validation set. We report AP, FLOPs, and number of parameters (Para). Methods are grouped by training epochs. Bold font indicates the best within each group.

Method	Backbone	AP	FLOPs	Para
<i>12-epoch training</i>				
DINO	R50	28.8	283G	48 M
RichSem	R50	32.2	283G	48 M
SIGMA	R50	36.4	303G	52 M
<i>24-epoch training</i>				
IGAM	R50-FPN	27.6	147G	44 M
H-Deformable DETR	R50	33.6	285G	49 M
DINO	R50	34.4	283G	48 M
RichSem	R50	34.9	283G	48 M
SIGMA	R50	39.6	303G	52 M

It can be observed that SIGMA attains the best AP among all compared methods under both training schedules, reaching 36.4 AP with 12-epoch training and 39.6 AP with 24-epoch training. Although SIGMA adopts a dual-branch architecture and further introduces query initialization and refinement modules, these additional components lead to only a moderate increase in FLOPs and parameters.

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In particular, compared with the single-branch baselines DINO and RichSem, SIGMA increases model complexity only from 283G FLOPs and 48M parameters to 303G FLOPs and 52M parameters, while yielding substantial AP gains. Moreover, compared with H-Deformable DETR, which is also based on a dual-branch design, SIGMA still achieves significant improvements under 24-epoch training, while introducing moderate extra complexity. Therefore, the proposed auxiliary branch and query enhancement strategy are effective in improving detection performance while keeping the additional training burden well controlled.